Reinforcement Learning for Portfolio Risk Management

CVaR-Constrained Dynamic Allocation

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Abstract

An applied treatment of reinforcement learning for dynamic portfolio allocation under tail-risk constraints. We formulate the allocation problem as a Constrained Markov Decision Process (CMDP) where the agent maximises risk-adjusted returns subject to a Conditional Value-at-Risk (CVaR) budget. Two optimisation paradigms are developed: policy-gradient (REINFORCE) and derivative-free Natural Evolution Strategies (NES). Reward shaping incorporates rolling CVaR penalties and transaction costs to produce a realistic trading signal. The RL agent is benchmarked against equal-weight, mean-variance, and risk-parity strategies, with the goal of achieving competitive returns while maintaining tighter control over drawdowns and tail risk. The document provides the mathematical foundations required to implement and interpret the code in Project 13.

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1 Portfolio Allocation as a Sequential Decision

Read Before Coding

This section assumes familiarity with basic portfolio theory (mean-variance optimisation) and Markov Decision Processes. We use CVaR as the central risk measure; see Project 05 (EVT & Tail Risk) for a self-contained treatment.

1.1 The Static Paradigm and Its Limits

The classical framework of Markowitz (1952) solves a *single-period* problem:

$$\max_{\boldsymbol{w}} \; \boldsymbol{w}^\top \boldsymbol{\mu} - \frac{\gamma}{2} \boldsymbol{w}^\top \boldsymbol{\Sigma}\, \boldsymbol{w}, \qquad \text{s.t.} \quad \mathbf{1}^\top \boldsymbol{w} = 1, \quad w_i \geq 0, \tag{1}$$

where $\boldsymbol{w} \in \mathbb{R}^N$ are portfolio weights, $\boldsymbol{\mu}$ is the vector of expected returns, $\boldsymbol{\Sigma}$ is the covariance matrix, and $\gamma > 0$ is a risk-aversion parameter. The solution is computed once and held until the next rebalancing date.

This approach has two fundamental problems:

(i) **Stationarity assumption.** The inputs $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ are treated as fixed, yet financial markets exhibit regime changes, time-varying volatility, and momentum effects.

(ii) **No adaptation.** A portfolio optimised for calm markets becomes dangerously concentrated when correlations spike during crises.

Core Idea

The core insight motivating this project: portfolio allocation is not a single optimisation problem but a *sequential decision process* where the agent must adapt to changing market conditions at every rebalancing step.

1.2 MDP Formulation

We model portfolio rebalancing as a discrete-time Markov Decision Process $(\mathcal{S}, \mathcal{A}, P, R, \gamma_d)$:

Definition 1.1 (Portfolio MDP)**.** The components are:

- **State** $s_t \in \mathcal{S}$: a feature vector encoding the agent's information at time t,

$$s_t = \big(\boldsymbol{r}_{t-L:t},\ \boldsymbol{w}_t^-,\ \sigma_t^{\text{rolling}},\ \text{DD}_t\big),$$

  where $\boldsymbol{r}_{t-L:t}$ are the most recent L asset returns, $\boldsymbol{w}_t^-$ are the current (pre-rebalance) portfolio weights, $\sigma_t^{\text{rolling}}$ is a rolling volatility estimate, and DD_t is the current drawdown from peak.

- **Action** $a_t \in \mathcal{A}$: the new portfolio weight vector $\boldsymbol{w}_t^+$ satisfying $\mathbf{1}^\top \boldsymbol{w}_t^+ = 1$ and $w_{t,i}^+ \geq 0$ (long-only).

- **Transition** $P(s_{t+1} \mid s_t, a_t)$: determined by market dynamics (asset returns), which the agent does not control.

- **Reward** $r_t = R(s_t, a_t, s_{t+1})$: a risk-adjusted portfolio return (detailed in Section 4).

- **Discount factor** $\gamma_d \in (0, 1]$: controls the agent's planning horizon.

The agent learns a *policy* $\pi_\theta(a \mid s)$ parameterised by θ that maps observed market conditions to portfolio allocations. The objective is:

$$\max_\theta \; J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T-1} \gamma_d^t \, r_t\right]. \tag{2}$$

Remark 1.1. Unlike supervised learning, there is no "correct" label for the optimal allocation. The agent discovers good strategies through *interaction* with the environment (historical or simulated market data), guided only by the reward signal.

2 CVaR-Constrained MDP

2.1 Why Constrain Tail Risk?

Maximising expected return (even risk-adjusted return) can still produce unacceptable tail losses. A portfolio with a high Sharpe ratio may simultaneously have catastrophic drawdowns during rare events. We want a *hard constraint* on how bad the worst outcomes can be.

Definition 2.1 (CVaR at level α)**.** For a loss random variable L and confidence level $\alpha \in (0, 1)$,

$$\mathrm{VaR}_\alpha(L) = \inf\{l : \mathbb{P}(L \le l) \ge \alpha\},$$

$$\mathrm{CVaR}_\alpha(L) = \frac{1}{1-\alpha}\int_\alpha^1 \mathrm{VaR}_u(L)\, du = \mathbb{E}\big[L \mid L \ge \mathrm{VaR}_\alpha(L)\big].$$

CVaR is the expected loss in the worst $(1-\alpha)$ fraction of scenarios. It is a *coherent* risk measure: subadditive, positive homogeneous, translation invariant, and monotone.

2.2 The Constrained Optimisation Problem

We augment the RL objective (2) with a CVaR constraint:

$$\max_\theta \; J(\theta) \qquad \text{subject to} \qquad \mathrm{CVaR}_\alpha\big(L^{\pi_\theta}\big) \le \bar{c}, \tag{3}$$

where L^{π_θ} is the portfolio loss distribution induced by policy π_θ and $\bar{c}$ is the CVaR budget (threshold).

2.3 Lagrangian Relaxation

Direct enforcement of (3) is intractable in the RL loop. We relax it via a Lagrangian:

$$\mathcal{L}(\theta, \lambda) = J(\theta) - \lambda \max\big(\mathrm{CVaR}_\alpha(L^{\pi_\theta}) - \bar{c},\; 0\big), \tag{4}$$

where $\lambda \ge 0$ is the Lagrange multiplier.

Key Insight

The multiplier λ is the **shadow price of risk**. It encodes how aggressively the agent trades return for tail-risk reduction:

- When $\mathrm{CVaR}_\alpha < \bar{c}$: the constraint is *inactive*, and the agent maximises return freely (λ contribution is zero).
- When $\mathrm{CVaR}_\alpha > \bar{c}$: the penalty activates, pushing the agent toward more conservative allocations.

A higher λ produces a more risk-averse agent; a lower λ tolerates larger tail losses in exchange for higher expected returns.

2.4 Dual Ascent on λ

In practice λ can be treated as a fixed hyperparameter or updated via dual gradient ascent:

$$\lambda_{k+1} = \max\bigl(0,\ \lambda_k + \eta_\lambda\,(\widehat{\mathrm{CVaR}}_\alpha - \bar{c})\bigr), \tag{5}$$

where $\widehat{\mathrm{CVaR}}_\alpha$ is the empirical CVaR estimated from recent rollouts and η_λ is the dual learning rate. This drives λ upward when the constraint is violated and downward when there is slack.

Warning

Dual ascent introduces an *inner–outer* loop: the inner loop updates θ (policy parameters) for a fixed λ; the outer loop updates λ based on whether the CVaR constraint is satisfied. Convergence requires the inner loop to approximately solve its sub-problem before each λ update. In practice, updating λ every K policy steps (with K between 5 and 20) works well.

3 Policy Gradient and Evolutionary Strategies

We consider two families of algorithms to optimise θ.

3.1 REINFORCE (Policy Gradient)

Given a parameterised stochastic policy $\pi_\theta(a \mid s)$, the policy gradient theorem (Williams, 1992) gives:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t \mid s_t)\,\hat{A}_t\right], \tag{6}$$

where $\hat{A}_t$ is an advantage estimate (e.g. return-to-go minus a baseline). The gradient is estimated by sampling trajectories and computing the log-probability-weighted returns.

REINFORCE for Portfolio Allocation

1. Sample M episodes $\{\tau^{(m)}\}_{m=1}^{M}$ under π_θ.
2. For each episode, compute returns $G_t^{(m)} = \sum_{k=t}^{T-1} \gamma_d^{k-t} r_k^{(m)}$.
3. Estimate gradient: $\hat{g} = \frac{1}{M}\sum_{m=1}^{M}\sum_t \nabla_\theta \log \pi_\theta(a_t^{(m)} \mid s_t^{(m)})\,(G_t^{(m)} - b)$.
4. Update: $\theta \leftarrow \theta + \eta\,\hat{g}$.

Pros: Principled gradient estimate; works with any differentiable policy (neural network with softmax output for weights).

Cons: High variance; requires careful baseline tuning; the CVaR penalty involves sorting (non-differentiable), which complicates gradient computation.

3.2 Natural Evolution Strategies (NES)

NES (Salimans et al., 2017) is a *derivative-free* black-box optimiser. Instead of computing gradients through the policy, it maintains a search distribution over parameters $\theta \sim \mathcal{N}(\boldsymbol{\mu}, \sigma^2 I)$ and uses fitness evaluations to estimate the natural gradient.

Definition 3.1 (NES Update)**.** Given a population of N_p perturbations $\{\boldsymbol{\epsilon}_i\}_{i=1}^{N_p}$, $\boldsymbol{\epsilon}_i \sim \mathcal{N}(\mathbf{0}, I)$, evaluate the fitness $F(\boldsymbol{\mu} + \sigma\,\boldsymbol{\epsilon}_i)$ for each perturbation. The parameter update is:

$$\boldsymbol{\mu} \leftarrow \boldsymbol{\mu} + \frac{\eta}{N_p\,\sigma} \sum_{i=1}^{N_p} F_i\,\boldsymbol{\epsilon}_i, \tag{7}$$

where $F_i = F(\boldsymbol{\mu} + \sigma\,\boldsymbol{\epsilon}_i)$ is the fitness of the i-th perturbation.

Mirror sampling. For each $\boldsymbol{\epsilon}_i$, also evaluate $-\boldsymbol{\epsilon}_i$, producing an antithetic pair. This halves the variance of the gradient estimate at the cost of doubling evaluations (a worthwhile trade in practice).

NES with Mirror Sampling for CVaR-Constrained Allocation

1. Initialise $\boldsymbol{\mu}$, perturbation scale σ, learning rate η, CVaR multiplier λ.
2. For each generation $k = 1, 2, \ldots$:
   (a) Sample $N_p/2$ perturbations $\boldsymbol{\epsilon}_i \sim \mathcal{N}(\mathbf{0}, I)$.
   (b) Evaluate fitness for $\boldsymbol{\mu} + \sigma\boldsymbol{\epsilon}_i$ and $\boldsymbol{\mu} - \sigma\boldsymbol{\epsilon}_i$ (mirror sampling).
   (c) Fitness = Lagrangian (4): episode return minus $\lambda \cdot \max(\widehat{\mathrm{CVaR}} - \bar{c}, 0)$.
   (d) Update $\boldsymbol{\mu}$ via (7).
   (e) Optionally update λ via (5).

Core Idea

Why NES for risk-constrained allocation?

1. **Non-differentiable objectives.** CVaR computation involves sorting returns and taking the tail mean — operations that break standard backpropagation. NES treats the objective as a black box.
2. **Robustness to noisy rewards.** Financial returns are inherently noisy. NES aggregates over a population of perturbations, smoothing out noise.
3. **Parallelisable.** Each perturbation can be evaluated independently, making NES well-suited for GPU or multi-core execution.

This is the same optimisation strategy used in Project 08 (Deep Hedging) for derivative-free policy search.

4 Reward Shaping for Risk

The reward signal is the mechanism through which we communicate our risk–return preferences to the agent. A poorly designed reward yields an agent that maximises the wrong quantity.

4.1 Composite Reward

At each rebalancing step t, the reward is:

$$r_t = \underbrace{R_t^{\text{port}}}_{\text{portfolio return}} - \underbrace{\lambda \max\big(\widehat{\mathrm{CVaR}}_{\alpha,t} - \bar{c},\ 0\big)}_{\text{CVaR penalty}} - \underbrace{\kappa \sum_{i=1}^{N} \big|w_{t,i}^{+} - w_{t,i}^{-}\big| \cdot c_{\text{rate}}}_{\text{transaction cost}}, \tag{8}$$

where:

- $R_t^{\text{port}} = \boldsymbol{w}_t^{+\top} \boldsymbol{r}_t$ is the realised one-step portfolio return.
- $\widehat{\text{CVaR}}_{\alpha,t}$ is the empirical CVaR computed from a rolling window of recent portfolio returns (e.g. the past 60 trading days).
- $\lambda \geq 0$ is the risk penalty weight (Lagrange multiplier).
- $\kappa \geq 0$ scales the importance of trading costs.
- c_{rate} is the per-unit proportional transaction cost (e.g. 10 basis points).

4.2 Effect of Hyperparameters

Parameter	Low value	High value
λ (risk penalty)	Aggressive, higher returns, larger drawdowns	Conservative, lower returns, tighter ta
κ (cost penalty)	Frequent rebalancing, responsive	Infrequent rebalancing, stable
$\bar{c}$ (CVaR threshold)	Tight constraint, limits tail risk early	Loose constraint, allows more risk
α (CVaR confidence)	Shallow tail ($\alpha = 0.90$), moderate risk	Deep tail ($\alpha = 0.99$), extreme risk

Warning

The CVaR penalty and the portfolio return operate on very different scales. If λ is too large relative to typical returns, the agent learns to hold all-cash or all-bonds. If λ is too small, the CVaR constraint has no effect. In practice, normalise returns and CVaR to similar magnitudes before tuning λ.

4.3 Rolling CVaR Estimation

Given a window of W recent portfolio returns $\{R_{t-W+1}^{\text{port}}, \ldots, R_t^{\text{port}}\}$:

1. Convert returns to losses: $l_k = -R_k^{\text{port}}$.
2. Sort losses in descending order: $l_{(1)} \geq l_{(2)} \geq \cdots \geq l_{(W)}$.
3. The number of tail observations: $n_\alpha = \lceil (1-\alpha)\, W \rceil$.
4. Empirical CVaR: $\widehat{\text{CVaR}}_\alpha = \frac{1}{n_\alpha} \sum_{k=1}^{n_\alpha} l_{(k)}$.

Remark 4.1. This is a non-parametric estimate. For small windows ($W < 100$) and high α (≥ 0.95), the estimate is noisy because only a few observations fall in the tail. Using $\alpha = 0.95$ with $W = 60$ yields approximately $n_\alpha = 3$ tail observations. This noise is part of why derivative-free optimisation (NES) is preferred over gradient methods.

5 Benchmark Strategies

To evaluate the RL agent, we compare it against three classical strategies that span the spectrum from estimation-free to model-dependent.

5.1 Equal Weight (1/N)

The simplest possible allocation:

$$w_i = \frac{1}{N}, \qquad i = 1, \ldots, N. \tag{9}$$

DeMiguel, Garlappi & Uppal (2009) showed that $1/N$ outperforms many "optimal" strategies in out-of-sample tests. The reason is instructive: optimal strategies introduce *estimation error* through $(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}})$, and this error often exceeds the benefit of optimisation.

Key Insight

$1/N$ is the natural baseline: any strategy that cannot beat equal weighting is not worth its complexity. Maximum diversification with zero estimation error.

5.2 Mean-Variance (Markowitz)

The tangency portfolio weights (assuming a risk-free rate of zero):

$$\boldsymbol{w}^{\mathrm{MV}} = \frac{\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}{\mathbf{1}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}. \tag{10}$$

This is optimal under the assumption that returns are i.i.d. multivariate normal and that $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ are known exactly. Neither assumption holds in practice:

- Returns are fat-tailed and exhibit volatility clustering.
- Estimated $\hat{\boldsymbol{\mu}}$ is notoriously unreliable; $\hat{\boldsymbol{\Sigma}}$ is more stable but still subject to regime changes.

Warning

Mean-variance portfolios are highly sensitive to the expected return vector $\boldsymbol{\mu}$. Small perturbations in $\hat{\boldsymbol{\mu}}$ can flip the optimal weights entirely. This is the *Markowitz curse*: the assets with the highest estimated expected returns also tend to have the highest estimation error, leading to extreme and unstable allocations.

5.3 Risk Parity

Risk parity sets weights so that each asset contributes equally to portfolio *volatility*:

$$w_i \propto \frac{1}{\sigma_i}, \qquad \text{normalised so that } \sum_i w_i = 1, \tag{11}$$

where $\sigma_i = \sqrt{\Sigma_{ii}}$ is the marginal volatility of asset i. The intuition: low-volatility assets receive higher weight, preventing the portfolio from being dominated by a single volatile asset.

Remark 5.1. Risk parity equalises *volatility* contributions, not *tail risk* contributions. During crises, correlations spike and even a volatility-balanced portfolio can suffer concentrated tail losses. This motivates the tail risk parity approach developed in Section 6.

5.4 Summary of Benchmark Properties

Strategy	Estimation needed	Adapts to regimes	Controls tail risk
Equal weight ($1/N$)	None	No	No
Mean-variance	$\boldsymbol{\mu}, \boldsymbol{\Sigma}$	No	No
Risk parity	$\boldsymbol{\Sigma}$ only	Partially	No
RL agent	Learned	**Yes**	**Yes**

6 Dynamic Risk Budgeting

6.1 Marginal CVaR

The *marginal CVaR* of asset i measures how sensitive the portfolio CVaR is to a small change in weight w_i:

$$\mathrm{MCVaR}_i = \frac{\partial\ \mathrm{CVaR}_\alpha(\boldsymbol{w})}{\partial w_i} = \mathbb{E}\big[r_i \mid R^{\mathrm{port}} \leq -\mathrm{VaR}_\alpha\big], \tag{12}$$

where $R^{\mathrm{port}} = \boldsymbol{w}^\top \boldsymbol{r}$ and the expectation is taken over the joint return distribution, conditioned on portfolio returns being in the worst $(1-\alpha)$ tail.

Definition 6.1 (CVaR Contribution)**.** The CVaR contribution of asset i is the product of its weight and its marginal CVaR:

$$\mathrm{CCVaR}_i = w_i \cdot \mathrm{MCVaR}_i.$$

By Euler's theorem for homogeneous functions, the total portfolio CVaR decomposes as:

$$\mathrm{CVaR}_\alpha(\boldsymbol{w}) = \sum_{i=1}^{N} \mathrm{CCVaR}_i = \sum_{i=1}^{N} w_i\, \mathrm{MCVaR}_i. \tag{13}$$

6.2 Tail Risk Parity

Definition 6.2 (Tail Risk Parity)**.** A portfolio satisfies tail risk parity if every asset contributes equally to portfolio CVaR:

$$\mathrm{CCVaR}_1 = \mathrm{CCVaR}_2 = \cdots = \mathrm{CCVaR}_N = \frac{\mathrm{CVaR}_\alpha(\boldsymbol{w})}{N}. \tag{14}$$

While standard risk parity equalises volatility contributions, tail risk parity equalises *tail* contributions. This is the risk-aware analogue: it prevents any single asset from dominating the portfolio's worst-case losses.

Core Idea

A well-trained RL agent should **approximately** converge to a tail risk parity allocation — not because it is told to, but because the CVaR penalty in the reward function incentivises equalising tail risk contributions. If an asset dominates the tail, the agent learns to underweight it. We validate this by comparing the agent's learned allocations with the analytical tail risk parity solution.

6.3 Empirical Marginal CVaR Estimation

Given M historical scenarios (or simulated paths) of asset returns $\{\boldsymbol{r}^{(m)}\}_{m=1}^{M}$:

1. Compute portfolio returns: $R^{(m)} = \boldsymbol{w}^\top \boldsymbol{r}^{(m)}$ for each scenario.
2. Identify the tail set: $\mathcal{T} = \{m : R^{(m)} \leq -\mathrm{VaR}_\alpha\}$, with $|\mathcal{T}| = \lceil (1-\alpha)M \rceil$.
3. Estimate marginal CVaR for each asset:
$$\widehat{\mathrm{MCVaR}}_i = -\frac{1}{|\mathcal{T}|} \sum_{m \in \mathcal{T}} r_i^{(m)}.$$
4. CVaR contribution: $\widehat{\mathrm{CCVaR}}_i = w_i \cdot \widehat{\mathrm{MCVaR}}_i$.

Remark 6.1. The RL agent does not compute marginal CVaR explicitly. Instead, it learns to avoid allocations that concentrate tail risk through the reward signal. The marginal CVaR analysis is a *diagnostic*: it lets us check whether the agent's behaviour is consistent with good risk budgeting.

7 Results Interpretation

Four diagnostic plots form the core of the evaluation framework for the RL agent.

7.1 Cumulative Return Curves

What it shows. The growth of \$1 invested under each strategy over the evaluation period.

What to look for.

- Does the RL agent *outperform* or *match* the benchmarks in total return?
- Is the growth path *smoother*? A strategy that earns the same total return with less volatility is superior.
- During *drawdown periods* (market crashes), does the RL agent lose less than the benchmarks?

Key Insight

The RL agent may not always have the highest terminal wealth. The goal is competitive returns with visibly better drawdown control. Look for the RL curve to "flatten" during crashes rather than plunging with the market.

7.2 Allocation Evolution Over Time

What it shows. Stacked area chart of portfolio weights $\boldsymbol{w}_t$ over time.

What to look for.

- **Regime adaptation**: does the agent shift toward defensive assets (bonds, low-volatility stocks) before or during market stress?
- **Smoothness**: are the allocation changes gradual (indicating stable learned features) or erratic (indicating overfitting to noise)?
- **Comparison with benchmarks**: equal weight is flat by construction, risk parity is nearly flat. The RL agent should show meaningful variation correlated with market conditions.

7.3 Risk–Return Scatter

What it shows. Each strategy plotted with annualised return on the y-axis and a risk metric (annualised volatility or CVaR) on the x-axis.

What to look for.

- A strategy in the *upper-left* quadrant (high return, low risk) dominates.
- The RL agent should sit *above* the line connecting equal weight and mean-variance, indicating a better risk–return trade-off.
- If using CVaR as the x-axis, the RL point should be near or below the threshold $\bar{c}$, confirming constraint satisfaction.

7.4 Drawdown Comparison

What it shows. Time series of drawdown $\mathrm{DD}_t = (V_{\max,t} - V_t)/V_{\max,t}$, where V_t is portfolio value and $V_{\max,t}$ is the running peak.

What to look for.

- **Maximum drawdown**: the deepest trough for each strategy. The RL agent should have a shallower maximum drawdown.
- **Recovery time**: how long does it take to return to the previous peak? The RL agent should recover faster.
- **Frequency**: does the RL agent experience fewer significant drawdown events overall?

Go Deeper

For a comprehensive treatment of drawdown-based risk measures, see Chekhlov, Uryasev & Zabarankin (2005), "Drawdown Measure in Portfolio Optimization." The maximum drawdown statistic is closely related to CVaR — both focus on tail events, but drawdown captures the temporal structure of losses.

8 References

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